CHATFIELD QUANTITATIVE RESEARCH

RESEARCH PROJECT SELECTION

ACADEMIC YEAR 2026/27

PRESENTED BY: **JAMES DAVIES**

Establishing our research direction, selecting our first economic forecasting project, and outlining our ambitions for the academic year.



ESTABLISH OUR STRATEGY

REVIEW METHODOLOGIES

EVALUATE APPLICATIONS

DISCUSS TERM 2 PROJECT

AGREE NEXT STEPS

OUR ANNUAL RESEARCH STRATEGY

TWO TERMS. TWO PROJECTS. TWO PUBLICATIONS.

BUILDING FOUNDATIONS AND AMBITION THROUGH QUANTITATIVE RESEARCH

TERM 1
FOUNDATIONS

ESTABLISHING OUR RESEARCH PROCESS

- Implement an established forecasting model
- Learn data collection and preparation
- Develop statistical programming skills
- Attempt manageable model improvements
- Establish team collaboration and research standards

PUBLISH RESEARCH ARTICLE 1
END OF TERM 1

TERM 2
ADVANCED RESEARCH

APPLYING OUR EXPERIENCE TO A LARGER INVESTIGATION

- Develop a more sophisticated forecasting system
- Apply our established research procedures
- Combine advanced statistical methodologies
- Investigate more ambitious model improvements
- Apply our collective technical experience

PUBLISH RESEARCH ARTICLE 2
END OF TERM 2

OUR GOAL | Produce two high-quality, reproducible quantitative research articles across the academic year.

OUR TERM 1 RESEARCH METHODOLOGY

FROM EXISTING RESEARCH TO ORIGINAL FINDINGS

USING ESTABLISHED METHODOLOGIES TO CONDUCT OUR FIRST INVESTIGATION

01 LITERATURE REVIEW

UNDERSTAND THE PUBLISHED RESEARCH

- Review the academic paper in detail
- Understand the methodology and economic context
- Identify key assumptions and limitations

02 DATA COLLECTION

SOURCE, CLEAN AND ORGANISE DATA

- Obtain relevant economic datasets
- Clean and preprocess the data
- Document data sources and transformations

03 MODEL IMPLEMENTATION

REPRODUCE AN ESTABLISHED APPROACH

- Implement the published statistical model
- Replicate key results where possible
- Validate our implementation

04 MODEL DEVELOPMENT

ATTEMPT A JUSTIFIED IMPROVEMENT

- Identify potential limitations
- Introduce additional indicators or methods
- Evaluate whether the modification improves forecasting performance

05 RESEARCH PUBLICATION

EVALUATE RESULTS AND WRITE OUR ARTICLE

- Compare model performance
- Explain and interpret our findings
- Produce a reproducible research article

EXPECTED OUTCOMES

- A functioning economic forecasting model
- Reproducible code and documented datasets
- A comparison between the baseline and modified model
- A published quantitative research article

KEY POINT

Our objective is not necessarily to outperform existing research, but to conduct a rigorous investigation into whether our proposed modifications improve forecasting performance.

SELECTING OUR FIRST RESEARCH PROJECT

TWO DECISIONS: ECONOMICS AND STATISTICS

EVERY PROJECT COMBINES AN ECONOMIC APPLICATION WITH A STATISTICAL METHODOLOGY

01 ECONOMIC APPLICATION

WHAT ECONOMIC VARIABLE OR RELATIONSHIP DO WE WANT TO INVESTIGATE?

COMMODITIES AND ENERGY MARKETS
e.g. natural gas, crude oil, energy prices

UK INFLATION AND PRICE MOVEMENTS
e.g. CPI, food prices, energy prices

LABOUR MARKET CONDITIONS
e.g. vacancies, earnings, employment

FOREIGN EXCHANGE MARKETS
e.g. GBP/USD exchange rates

+

RESEARCH PROJECT

A combination of an economic application and a statistical methodology

02 STATISTICAL METHODOLOGY

WHICH MATHEMATICAL APPROACH WILL WE IMPLEMENT?

FUNDAMENTAL REGRESSION FORECASTING
Model relationships between economic indicators and forecast the target variable.

AUTOREGRESSIVE / ARIMA FORECASTING
Use historical observations to forecast economic variables.

STOCHASTIC RANDOM-WALK FORECASTING
Model economic price movements probabilistically using estimated parameters.

OUR SELECTION PROCESS

REVIEW THE THREE MODELS → CONSIDER THREE APPLICATIONS FOR EACH MODEL → EVALUATE FEASIBILITY AND INTEREST → DISCUSS AS A TEAM → SELECT THE MOST SUITABLE COMBINATION

MODEL 1: FUNDAMENTAL REGRESSION FORECASTING

USING ECONOMIC FUNDAMENTALS TO IMPROVE FORECASTS

A REGRESSION-BASED APPROACH TO MODEL ECONOMIC RELATIONSHIPS

1 REFERENCE PAPER

Driving Fundamentals of Natural Gas Price in Europe (2020)

International Journal of Energy Economics and Policy

View paper (DOI: 10.32479/ijeep.10192)

2 METHODOLOGY

Use regression to estimate the relationship between economic indicators and the variable we want to forecast, and investigate whether additional indicators improve forecasting performance.

3 IMPLEMENTATION

- Construct a baseline regression model using historical economic data, following a simplified version of the published methodology.
- Estimate model parameters and generate out-of-sample forecasts.

4 PROPOSED IMPROVEMENT

- Introduce additional economic indicators and test whether they improve out-of-sample forecasting accuracy.
- Compare the baseline and extended models using appropriate forecast evaluation metrics.

5 POTENTIAL ECONOMIC APPLICATIONS



A EUROPEAN NATURAL GAS

Can European gas storage levels improve forecasts of Dutch TTF natural gas prices?



B UK INFLATION

Can energy prices and interest rates improve forecasts of UK inflation?



C BRENT CRUDE OIL

Can oil inventories and production indicators improve forecasts of Brent crude oil prices?

RESEARCH OBJECTIVE

Investigate whether incorporating relevant economic fundamentals improves forecasting performance relative to a simple baseline model.

SKILLS DEVELOPED

- Regression modelling
- Parameter estimation
- Economic interpretation
- Data preparation
- Forecasting evaluation

MODEL 2: AUTOREGRESSIVE / ARIMA FORECASTING

USING HISTORICAL DATA AND ALTERNATIVE INDICATORS

A TIME-SERIES APPROACH TO IMPROVE ECONOMIC FORECASTING

1 REFERENCE PAPER

Evaluating a New Earnings Indicator (2020)

ESCoE Discussion Paper Series No. 2020-19

View paper (ESCoE, 2020)

2 METHODOLOGY

Use historical economic observations to construct an autoregressive or ARIMA model, and investigate whether incorporating alternative data sources improves forecasting accuracy.

3 IMPLEMENTATION

- Fit an autoregressive or ARIMA model using historical observations of the economic variable.
- Extend to include alternative indicators through dynamic regression or ARIMAX modelling.
- Generate and evaluate out-of-sample forecasts.

4 PROPOSED IMPROVEMENT

- Incorporate alternative economic data sources (e.g. online job adverts, supermarket prices).
- Assess whether these indicators provide additional forecasting information beyond historical observations alone.

5 POTENTIAL ECONOMIC APPLICATIONS



A UK LABOUR-MARKET VACANCIES

Can Indeed job postings improve forecasts of official ONS job vacancies?



B UK FOOD INFLATION

Can supermarket price data provide earlier indications of official UK food inflation?



C UK EARNINGS GROWTH

Can advertised salaries from online job listings improve forecasts of official UK earnings growth?

RESEARCH OBJECTIVE

Investigate whether alternative economic indicators provide additional forecasting information beyond historical observations alone.

SKILLS DEVELOPED

- Time-series forecasting
- Autoregressive modelling
- Alternative economic data
- Data alignment and cleaning
- Model evaluation
- Economic interpretation

MODEL 3: STOCHASTIC RANDOM-WALK FORECASTING

MODELLING ECONOMIC PRICE MOVEMENTS PROBABILISTICALLY

A STOCHASTIC PROCESS APPROACH TO GENERATE AND EVALUATE PROBABILISTIC FORECASTS

1 REFERENCE PAPER

Forecasting the Nominal Brent Oil Price with VARs—One Model Fits All? (2015)

International Monetary Fund Working Paper 15/251

View paper (IMF, 2015)

2 METHODOLOGY

Model economic price movements as a stochastic process, estimating key parameters from historical data and generating probabilistic forecasts using Monte Carlo simulation.

3 IMPLEMENTATION

- Implement a random walk with drift model for the chosen economic variable.
- Estimate drift and volatility parameters from historical data using maximum likelihood.
- Generate probabilistic forecasts using Monte Carlo simulation.

4 PROPOSED IMPROVEMENT

- Introduce rolling-window volatility estimation to allow parameters to adapt to recent market conditions.
- Investigate whether adaptive volatility estimates improve forecast uncertainty calibration and out-of-sample performance.

5 POTENTIAL ECONOMIC APPLICATIONS



A BRENT CRUDE OIL

Can stochastic models generate reliable probabilistic forecasts of Brent crude oil prices?



B EUROPEAN NATURAL GAS

Can stochastic models capture uncertainty in Dutch TTF natural gas prices?



C GBP/USD EXCHANGE RATE

Can a random-walk model generate well-calibrated probabilistic forecasts of GBP/USD exchange rates?

RESEARCH OBJECTIVE

Investigate whether adapting stochastic model parameters to recent market conditions improves the reliability and calibration of probabilistic forecasts relative to a simple baseline model.

SKILLS DEVELOPED

- Stochastic processes
- Maximum likelihood estimation
- Monte Carlo simulation
- Volatility estimation
- Probabilistic forecasting
- Model evaluation

TERM 1 RESEARCH PROJECT SELECTION

COMPARING THE THREE FORECASTING MODELS

KEY FEATURES, STRENGTHS AND CONSIDERATIONS FOR OUR FIRST INVESTIGATION

FEATURE	1 FUNDAMENTAL REGRESSION FORECASTING	2 AUTOREGRESSIVE / ARIMA FORECASTING	3 STOCHASTIC RANDOM-WALK FORECASTING
			
Core approach	Use regression models to relate economic indicators to the target variable.	Model the economic variable using its own historical values (and lags).	Model price movements as a stochastic process with probabilistic forecasts.
Key data requirements	Macroeconomic and market indicators (e.g. energy prices, inflation, interest rates).	Historical time-series data for the target variable.	Historical price or return data for the target variable.
Main strengths	Interpretability, established methodology, clear economic intuition.	Effective for capturing time-series dynamics and autocorrelation.	Accounts for uncertainty, flexible, suited to volatile markets.
Potential applications	European natural gas, UK inflation, labour market conditions, FX rates.	UK inflation, earnings growth, commodity prices, GDP indicators.	Brent crude oil, natural gas, foreign exchange rates.
Key considerations	Assumes relatively stable relationships; may not capture structural changes.	Relies on historical patterns; may struggle with structural breaks or regime changes.	More complex to implement and calibrate; requires careful parameter estimation and validation.

VOTING CRITERIA

1. Economic relevance and interest
2. Feasibility within Term 1 timeframe
3. Learning opportunities and technical challenge
4. Availability and quality of data
5. Potential for meaningful results
6. Team interest and skills alignment

VOTING PROCESS

Review the three models → Discuss as a team → Vote on preferred project

POTENTIAL EXTENSION PROJECT

CUSUM CHANGEPOINT DETECTION AND REINFORCEMENT LEARNING

A DYNAMIC, REGIME-AWARE APPROACH TO ECONOMIC FORECASTING

1 CUSUM CHANGEPOINT DETECTION

Detect structural breaks in economic time series using the CUSUM method, following Dr Gaetano Romano's research.

2 REINFORCEMENT LEARNING

Use detected regime changes as states in a reinforcement learning framework to adapt forecasting strategy dynamically.

3 ADAPTIVE FORECASTING

Combine changepoint detection and RL to produce a regime-aware forecasting model that adjusts to new market conditions.

INSPIRED BY

Dr Gaetano Romano's research on CUSUM changepoint detection and reinforcement learning for financial time series.

Recent papers (2024–2025)

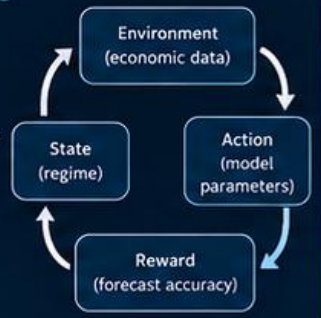
1 CUSUM CHANGEPOINT DETECTION

- Identify points in time where the statistical properties of the series change.
- Detect regime shifts in macroeconomic and financial indicators (e.g. inflation, interest rates).
- Provides an objective, data-driven method to separate different market regimes.



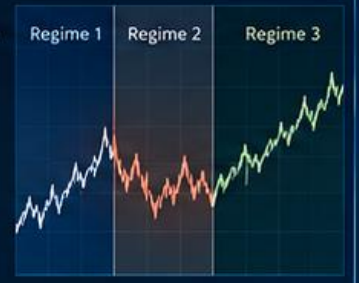
2 REINFORCEMENT LEARNING

- Treat each detected regime as a state in an RL environment.
- Learn the optimal forecasting strategy for each regime.
- Adapt to changing market conditions over time, rather than relying on a single static model.



3 ADAPTIVE FORECASTING

- Combine changepoint detection and RL to create a model that adapts to new regimes.
- Aims to improve forecast accuracy and robustness relative to standard models.
- Particularly relevant for volatile markets with frequent structural changes.



POTENTIAL ECONOMIC APPLICATIONS

A INFLATION FORECASTING

Capture structural changes in inflation dynamics (e.g. post-pandemic regime shifts).

B COMMODITY PRICE MODELLING

Adapt to regime changes in oil and natural gas markets.

C INTEREST RATE FORECASTING

Improve forecasts of central bank policy and yields by accounting for regime shifts.

KEY CONSIDERATIONS

- Choice of CUSUM parameters and detection thresholds.
- Definition of states and rewards in the RL framework.
- Availability and quality of high-frequency or long historical data.
- Computational complexity and model validation.
- Evaluate performance against standard regression, ARIMA and baseline models.

CONCLUSION AND NEXT STEPS

KEY TAKEAWAYS AND FUTURE DIRECTION

WHAT WE HAVE LEARNT, WHERE THIS COULD LEAD, AND HOW WE MOVE FORWARD

1 KEY TAKEAWAYS

Forecasting models rely on structure
Incorporating relevant economic fundamentals and alternative indicators can improve forecasting accuracy.

Model choice depends on the data
Different models capture different dynamics; there is no universally best approach.

Importance of robustness and validation
Out-of-sample testing, alternative specifications and probabilistic evaluation are essential.

Relevance to real-world decisions
Improved economic forecasts can inform policy, investment and risk management decisions across multiple markets.

2 MODEL COMPARISON SUMMARY

Aspect	Regression	AR / ARIMA	Random Walk
Main idea	Use economic fundamentals	Use historical patterns	Model price movements probabilistically
Key strengths	Interpretable, incorporates multiple indicators	Captures autocorrelation and time-series dynamics	Flexible, suitable for volatile markets
Key limitations	May miss structural breaks	Relies on past patterns; may struggle with regime changes	Requires careful parameter calibration
Best suited for	Relatively stable relationships	Markets with strong historical structure	Markets with high uncertainty and volatility

3 POTENTIAL FUTURE WORK

Additional economic indicators
Incorporate further macroeconomic and financial variables.

Regime-switching models
Explore Markov switching or time-varying parameter models.

Machine learning approaches
Compare with advanced ML models (e.g. LSTM, gradient boosting).

Broader market applications
Extend to other commodities, exchange rates and equity markets.

Real-time forecasting
Investigate online learning and adaptive methods for live forecasting.

4 OVERALL CONCLUSION

This project demonstrates that incorporating economic fundamentals and adaptive modelling techniques can **improve the accuracy and robustness of economic forecasts**, with important applications for policy, investment and risk management.



5 NEXT STEPS FOR OUR TEAM

1. DISCUSS
Review findings and discuss potential extensions.

2. PRIORITISE
Select 1–2 ideas for further investigation this term.

3. PLAN
Define data requirements, methodology and timeline.